Status	Finished
Started	Wednesday, 21 October 2026, 8:20 PM
Completed	Wednesday, 21 October 2026, 8:30 PM
Duration	9 mins 18 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It is slower than linear search for small lists
c. It requires the list to be sorted ✓
d. It does not work with negative numbers

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 90 and 94
c. 89 and 99
d. 90 and 99 ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Merge search
b. Linear search ✓
c. Bubble search
d. Quick search

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. In the entire list
c. At the beginning of the list
d. In the left sublist ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list
b. The search stops immediately ✓
c. The search starts over from the beginning
d. The search backtracks to find duplicate elements

Question **6**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By comparing each element in the list sequentially ✓
b. By sorting the list first
c. By dividing the list into halves
d. By using a hash function

Question **7**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. n ✓
c. 1
d. log n

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search is more efficient than binary search on large lists.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search requires the list to be sorted.

Question **9**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓
b. It compares elements sequentially
c. It works on unsorted lists
d. It always starts from the beginning of the list

Question **10**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Linear search ✓
b. Jump search
c. Binary search
d. Hashing

Question **11**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all
b. When the item is the last element in the array
c. Item is the last element in the array or item is not there at all
d. When the item is somewhere in the middle of the array ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements
b. When the element is found in the list ✓
c. When the list is sorted
d. When the search algorithm finishes

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The list is sorted
c. The search continues in the left or right sublist ✓
d. The search continues from the beginning

Question **14**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Bubble ✓

c. Insertion

d. Selection

Question **15**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search ends successfully ✓
b. The search continues in the left sublist
c. The search continues in the right sublist
d. The list is sorted